DSX AI Factory Platform
AI Inference Workshop

From first request to a resilient inference fleet, live on Kubernetes

Salt Lake City | Mon, Nov 9 | 09:00 – 12:00

Your morning at a glance

Mon, Nov 9 · 08:30 – 12:00 Mountain Time · Wasatch Ballroom, Radisson Hotel Salt Lake City Downtown

09:00
5 min
Welcome: what is DSX OS?
Bob Wise

09:05
5 min
Today's menu: the stack for the day
Ekin Karabulut

09:10
20 min
The foundation: NVKR, NVKE, AICR
Eric Chapman, Nathan Taber

09:30
lightning round
Slinky in two minutes
Tim Wickberg

09:32
≈ 2 h 20 min
Hands-on with NVIDIA Dynamo
Live build: one coding assistant on an open model, from first request to scale-out and recovery
Neal Vaidya

11:50
5 min
Coming next: Agent Substrate
Ekin Karabulut, Neal Vaidya

11:55
5 min
Action items, then lunch
Ekin Karabulut



Talk · Live demo

Questions welcome anytime.

09:00 · 5 MIN

What is DSX AI Factory Platform?

Bob Wise



Building Together, Growing the Community

Collaboration Matters - Today's Focus

- **DSX OS AI Inference:** Advance DSX together
- **Contributor Workshop**: Turn shared challenges into progress
- **Community Meetings**: Sustain momentum through community meetings



Bob Wise

VP Eng/Ops DGX Cloud

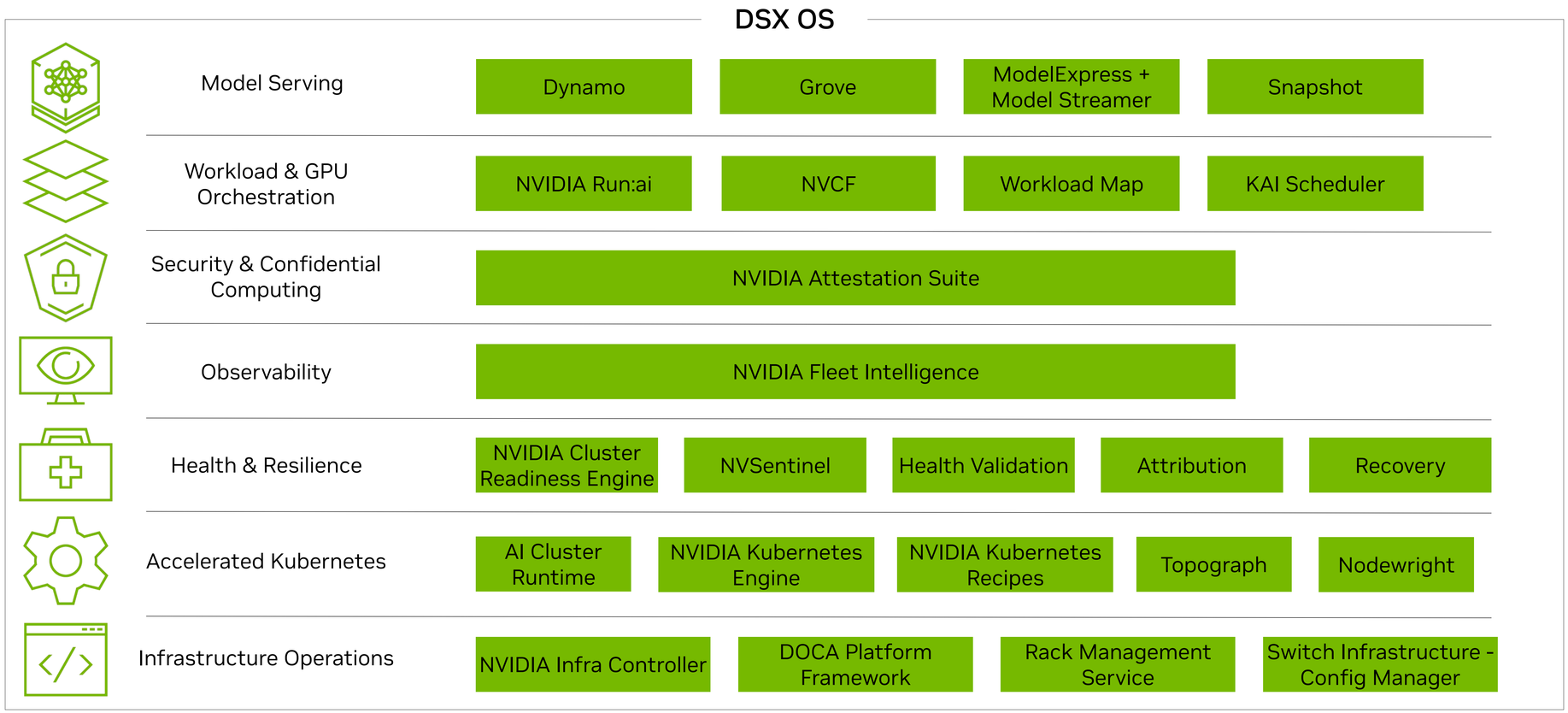
bwise@nvidia.com

Connect with Bob

Connect via LinkedIn

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NVIDIA DSX AI Factory Platform



DSX Sim

- Blueprints
- DSX Air
- SimReady

DSX MaxLPS

- Dynamic Power Software

DSX Exchange

- Event Bus
- Agent Gateway

DSX Flex

- Flexible Power Schema

09:05 · 5 MIN

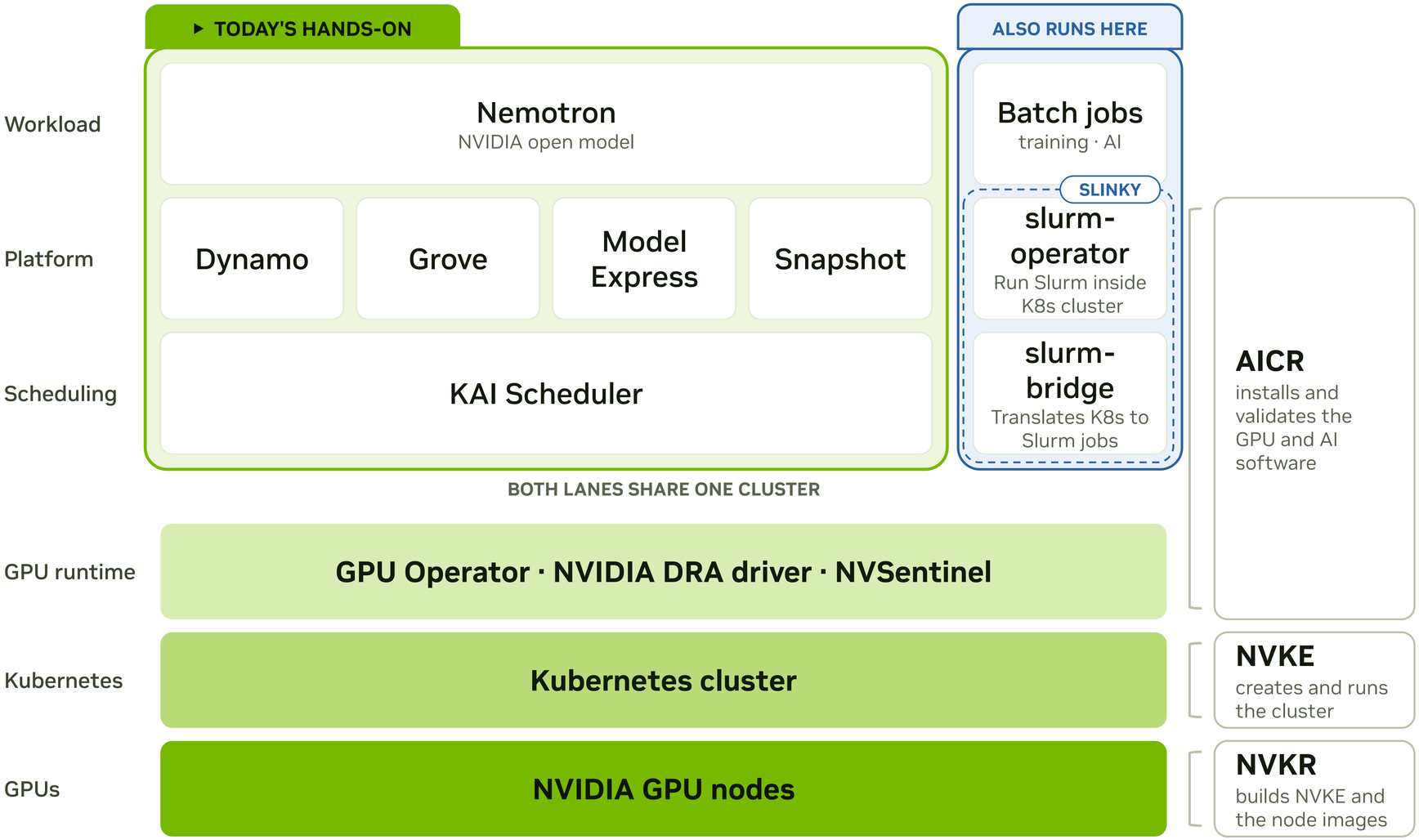
Today’s Menu

Ekin Karabulut



The stack for the day

From nodes to the model, in one stack



09:10 · 20 MIN

The Foundation: NVKR, NVKE, AICR

Eric Chapman, Nathan Taber



The NVIDIA Kubernetes Ecosystem

Cluster lifecycle management outside workload clusters, with AI runtime components inside

AI Workloads and Platforms

NEW!

NVKR
NVIDIA published recipes and build pipelines

Operator runs the build and publishes into their registry

AI Factory Registry

- Node OS Images
- Kubernetes packages
- AICR bundles

AICR
AI Cluster Runtime generates and validates the GPU and AI software configuration

Installs into any cluster

NEW!

NVKE
Creates and runs the Kubernetes cluster

NEW!

Cluster API Provider for NICO
Provision NICo bare-metal infrastructure via K8s

Kubernetes GPU cluster

Maximize GPU utilization

NEW!

NVIDIA Cluster Readiness Engine
Node testing and qualification

NVSentinel
Fault detection and remediation

Topograph
Topology discovery, active labeling

DRA Driver
GPU Resource allocation

KAI Scheduler
GPU-aware scheduling

OVN-Kubernetes
AI CNI with DPU offload

Configure GPU hardware in the cluster

Network Operator
Accelerated networking configuration

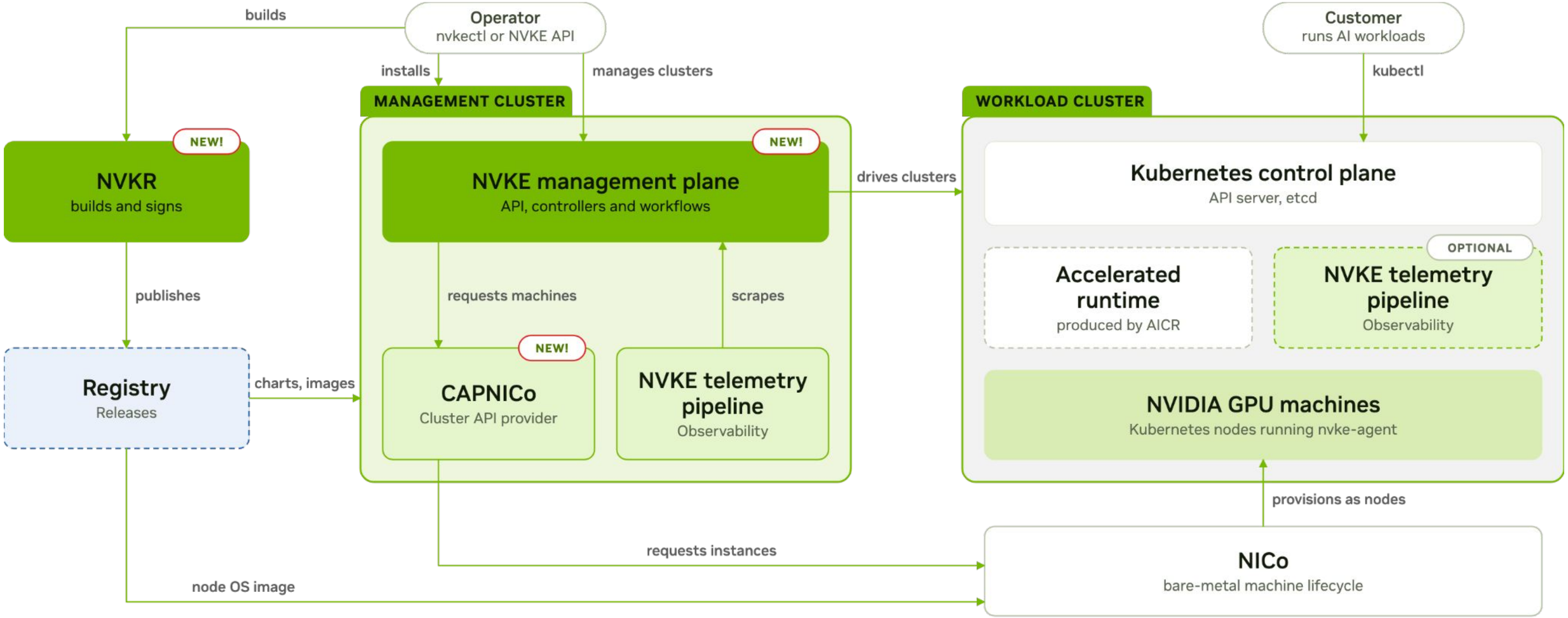
GPU Operator
GPU drivers and device software

Nodewright
Node OS and kernel configuration

NVIDIA GPU infrastructure

NVIDIA

Build the Kubernetes foundation with NVKE and NVKR



What is AICR?

AI Cluster Runtime (AICR): an open-source engine and CLI for reproducible GPU cluster configurations.

Optimized
Tuned for specific hardware, service, OS, and workload

Validation built in
Constraint checks and functional validation

Reproducible
Same release and inputs → same configuration

Composable
Layered overlays for public or private cloud

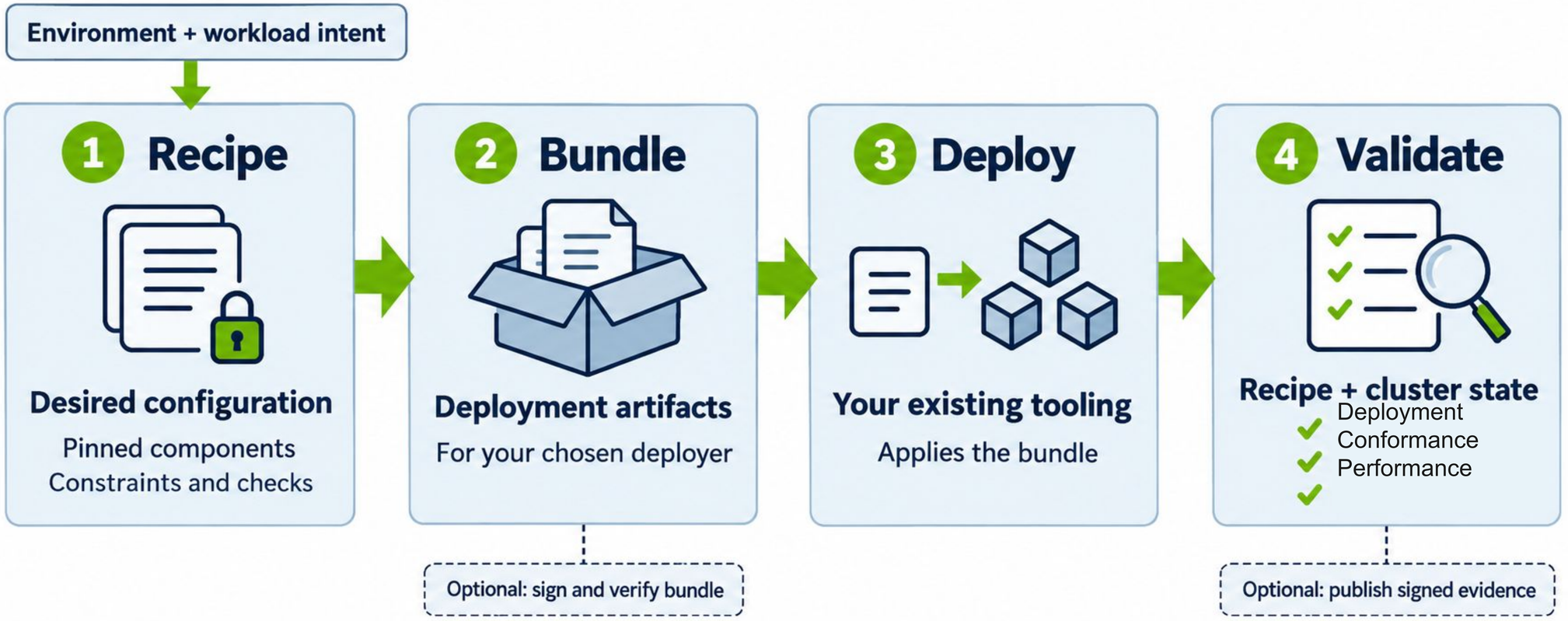
Verifiable
Release provenance, SBOMs, bundle signing

Fits your tooling
Helm, Argo CD, Flux, Terraform, and more

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AICR workflow

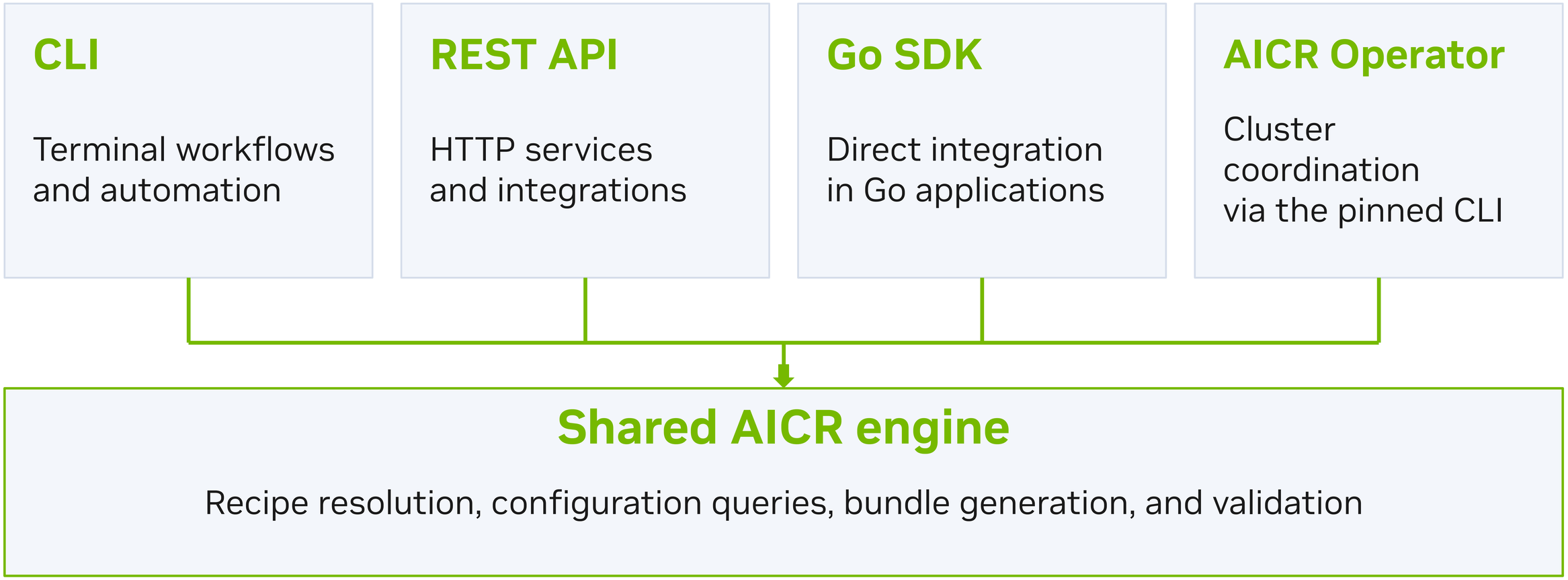
Use criteria or aicr snapshot for environment facts. Choose intent and an optional platform.



One recipe connects configuration, deployment, and validation.

Four ways to use AICR

Terminal, HTTP, Go, and Kubernetes workflows share the same AICR engine



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What’s supported today

Shipped overlay criteria · October 29, 2026 · AICR 4629f6d

12 Kubernetes services shown

Managed cloud	EKS · GKE · AKS · OKE · LKE · MK8s
Distributions	OpenShift · RKE2* · k0s* · NVIDIA NVKE*
Self-managed	Bare metal · kind (dev/test)

11 GPU accelerators

Rubin	VR200*
Blackwell	GB300 · GB200 · B300 · B200 · RTX PRO 6000
Hopper	H200 · H100
Ampere / Ada	A100 · L40S · L40

5 Operating systems

Ubuntu · COS · Oracle Linux · Amazon Linux · SLES

Talos (mixin; not counted)

6 Deployer families

Helm · Helmfile · Argo CD · Flux · Terraform · Fleet

5 Workload platforms

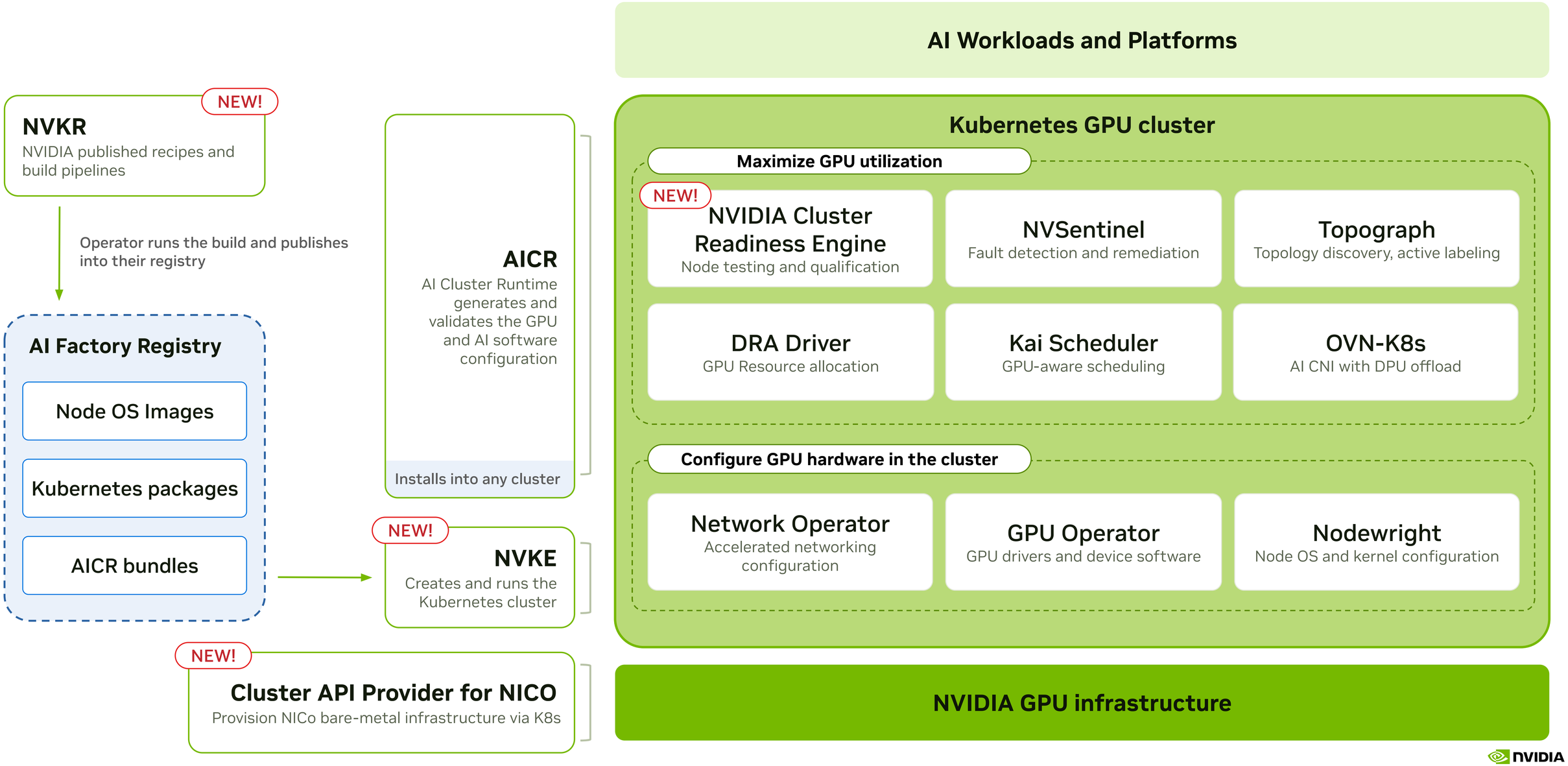
Training	Kubeflow · Slurm · Kueue
Inference	Dynamo · NIM

** Preview. Not every combination has a recipe. Argo CD has two output formats, so six deployer families have seven CLI formats.*

NVIDIA

How the NVIDIA Kubernetes Ecosystem Fits Together

Cluster lifecycle management outside workload clusters, with AI runtime components inside



09:30 · 2 MIN

Quick Overview: Slinky

Tim Wickberg



Slinky: Unified Scheduling of Slurm and Kubernetes

Bringing Slurm and Kubernetes together

Compute resources are limited.

User requests job → **Job gets prepared** → **Job in Queue** → **Jobs Executed**

Developers

Submit directly in Kubernetes using: **`kubectl`**

Pods get translated to Slurm jobs

slurm-bridge

Researchers

Submit using Slurm workflows: **`sbatch`**

Slurm batch jobs

Shared Scheduling of Slurm and K8s

Unified queue

- Priority + preemption
- Backfill
- Fair share
- Pod groups

Increased Access to Compute

Unified compute resources

Inference and training can coexist on the same resources

Bring together Slurm and Kubernetes: slurm + = slinky

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09:32 → 11:50 · LIVE BUILD

Let’s get started with NVIDIA Dynamo

Neal Vaidya



11:50 · 5 MIN

Coming next...

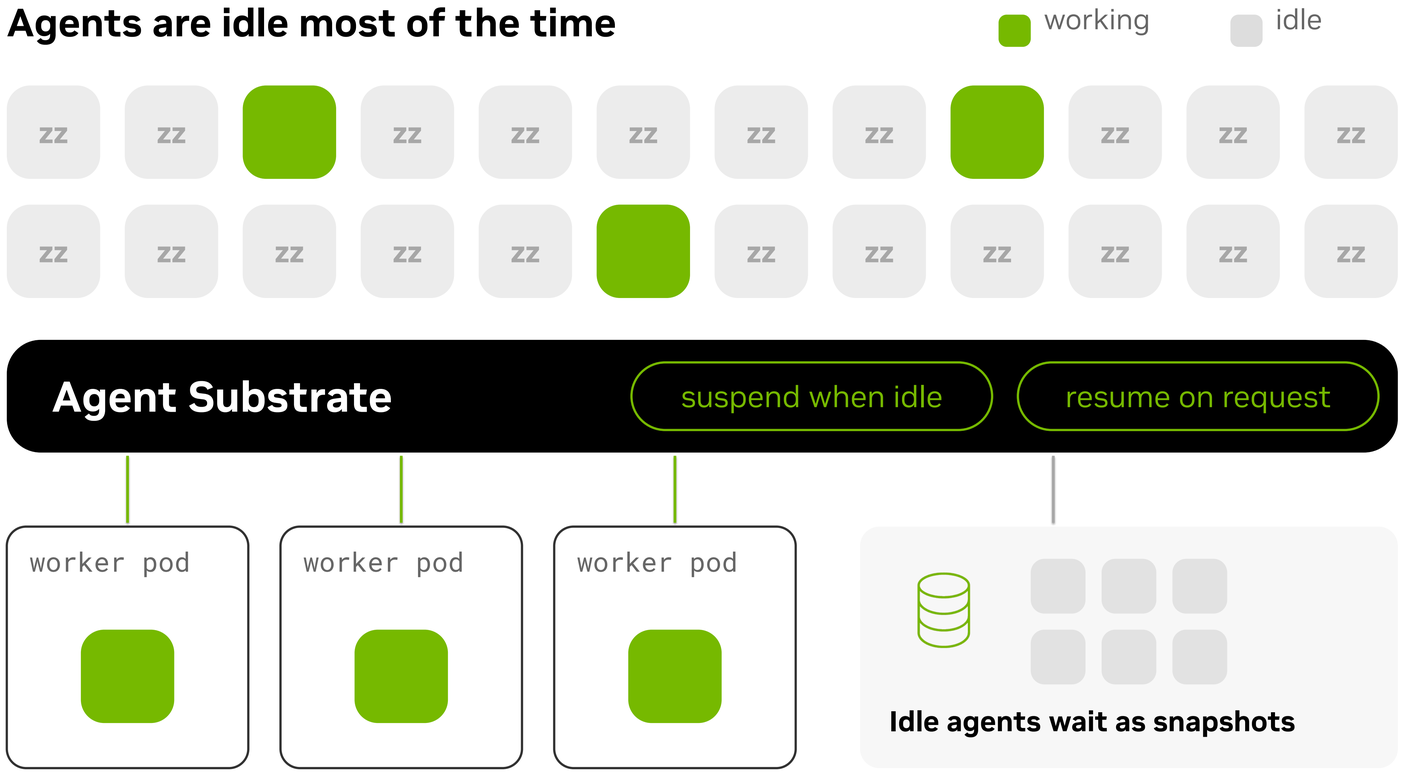
Ekin Karabulut & Neal Vaidya



Agent Substrate: millions of agents, a few warm workers

Zero-trust sandboxing for millions of agents

Agents are idle most of the time



A small pool of warm pods, scheduled by Kubernetes

Resumed onto any free worker

WHAT IT IS

Agent Substrate is a security-hardened, zero-trust runtime for sandboxed agents, engineered to safely run millions of agents and tools with 10x higher density than standard containers.

EVERY THURSDAY

Join the community meeting

10–11 AM PT · notes, recordings, Slack

WED, NOV 11 · 1:00 PM

Lightning talk: From Pods to Actors

Ron Lev · Google Cloud booth #122

Meet the Experts: Wed 3:15 PM, NVIDIA booth #170

NVIDIA

11:55 · 5 MIN

Action Items before Lunch

Ekin Karabulut



Scan, join, come say hi

Keep building with us

01

NVIDIA at KubeCon

Every NVIDIA session this week, plus booth #170

nvidia.com KubeCon page

02

Meet the Experts

20-minute drop-in Q&As with maintainers at booth #170, Tue to Thu

dsx-ai-factory/community

03

NVIDIA Dynamo Slack

Dynamo and its components, part of the DSX AI Factory

ai-dynamo.org/slack

04

DSX AI Factory Slack

All the other DSX AI Factory components and their teams

dsx-ai-factory.slack.com

05

CNCF Slack

#kai-scheduler: KAI Scheduler, resource management, GPU fractioning

#substrate-users: Agent Substrate

slack.cncf.io

12:00 **Lunch is next.** Hungry for more? Find us at NVIDIA booth #170 all week.

Released to Public

Not Yet Public

NVIDIA DSX OS

DSX OS



Model Serving

Dynamo | Grove | Model Streamer | Snapshot



Workload & GPU Orchestration

NVIDIA Run:ai | NVCF | Workload Map | KAI Scheduler



Security & Confidential Computing

Attestation Suite | Attestation Suite - Trust Outpost | Workload Identity



Observability

Fleet Intelligence | OTel Collector | Domain Analyzer | NVIDIA Observability



Health & Resilience

Cluster Readiness Engine | NVSentinel | Health Validation | Attribution | Recovery | Remediation Orchestrator



Accelerated Kubernetes

AI Cluster Runtime | NV Kubernetes Engine + Recipes | GPU Operator | Network Operator | Topograph | Nodewright



Infrastructure Operations

NVIDIA Infra Controller | DOCA Platform Framework | Rack Management Service | Switch Infrastructure - Config Manager | NVIDIA AI Infrastructure SDN | NVIDIA Infra Provisioning Engine

VMaaS | Control Plane Deployment Kit | Distributed Test Manager | Switch Infrastructure - Inventory Manager

DSX Sim



Stratum Sim | DSX Air

Blueprints | SimReady

DSX MaxLPS



Dynamic Power Software

DSX Exchange



Event Bus

Agent Gateway

DSX Flex



Flexible Power Schema

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